

Modeling Mayan Texts

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Stemma Codicum for the Popol Wuj



Precolonian
oral and inscribed

from 250 BCE



First composition
from codex (unseen)

~1545-58



Ximenez's
transcription

~1701-03

QUATRIÈME PARTIE.

CHAPITRE PREMIER.

Voilà donc que déjà beaucoup de villes s'étaient fondées, chacune à part (l'une de l'autre), et chacune des tribus se réunissait aux villes qui s'arrondissaient sur tous les chemins et leurs chemins étaient ouverts.

Quant à Balam-Quitzé, à Balam-Agab, à Mahucutah et à Iqui-Balam, on ne voyait pas clairement où ils étaient (1). Lorsqu'ils apercevaient (les gens) des tribus qui passaient par les chemins, aussitôt ils criaient au bord des monta-

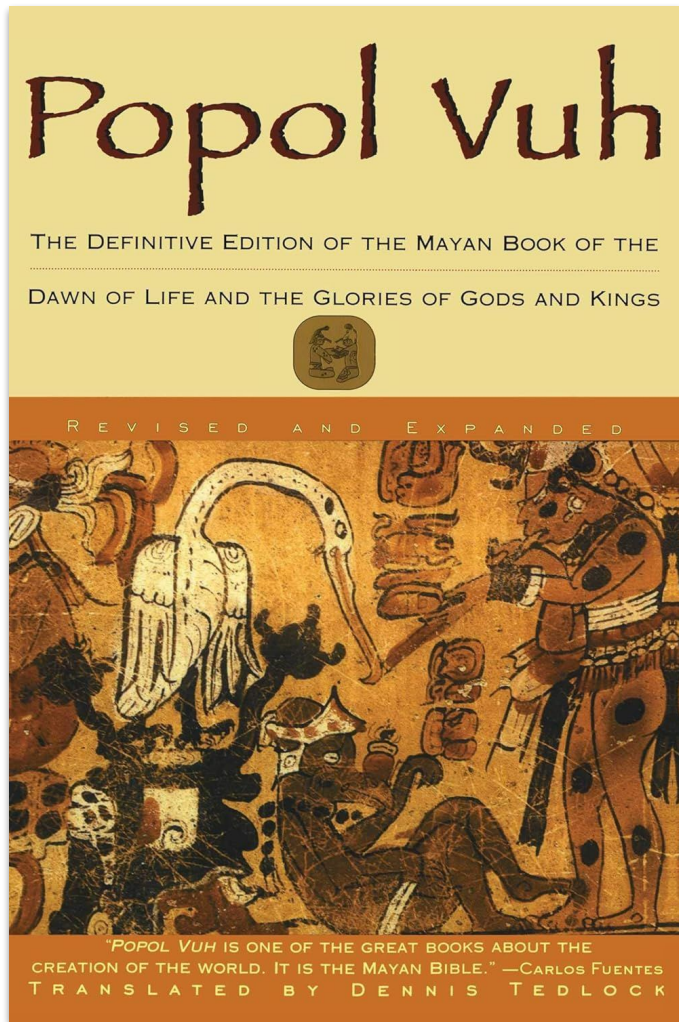
CAPITULO PRIMERO

AHORA bien, muchos pueblos fueron fundándose uno por uno, y las diferentes ramas de las tribus se iban reuniendo y agrupando junto a los caminos, sus caminos que habían abierto.

En cuanto a Balam-Quitze, Balam-Acab, Mahucutah e Iqui-Balam, no se sabía dónde estaban. Pero cuando veían a las tribus que pasaban por los caminos, al instante se ponían a gritar en la cumbre de los montes, lanzando el aullido del coyote y el grito del gato de monte, e imitando el rugido del león y del tigre.

Translated
editions

1857 to present



Is the *Popol Wuj* a single text?

Does it have a unified narrative structure?

A sample of editions shows **low consensus** in part and chapter divisions

ID	Author	Year	Language	Parts	Chapters	Comments
Sc	Scherzer	1857	Spanish	—	—	Follows original paragraphs.
Br	Brasseur (de Bourborg)	1861	French	4	45	Introduces divisions.
Re	Recinos	1947	Spanish	4	46	Preamble not included in parts.
Ed	Edmonson	1971	English	4	97	Follows original paragraphs.
Te	Tedlock	1985	English	5	80	
Ch	Christenson	2003	English	—	87	Follows original paragraphs.
Co	Colop	2011	Spanish	5	—	Colop calls parts chapters.
Ba	Bazzett	2018	English	4	44	Excludes historical chapters.
So	Sotelo	2018	Spanish	4	43	

Table 1: Editions and their divisions

For comparison, we use Ch 2003 chapters as a baseline

Table 2: Edition divisions compared to Ch 2003

Ch	Br	Re	Ed	Te	Co	Ba	So	Title
1	I		I	I	I	I	I	Preamble.
2		I						The Primordial World.
9			II		II	II		The Pride of Seven Macaw before the Dawn.
10				II				The Fall of Seven Macaw and his Sons.
15	II	II	III	III	III	III		The Tale of the Father of Hunahpu and Xbalanque.
22							II	Hunahpu and Xbalanque in the House of the Grandmother.
41	III	III	IV	IV	IV	IV	III	The Creation of Humanity.
57	IV*	IV*					IV	Offerings are Made to the Gods.
65				V	V			The Sons of the Progenitors Journey to Tulan.

However, **all agree** on dividing the text at **Ch 41**

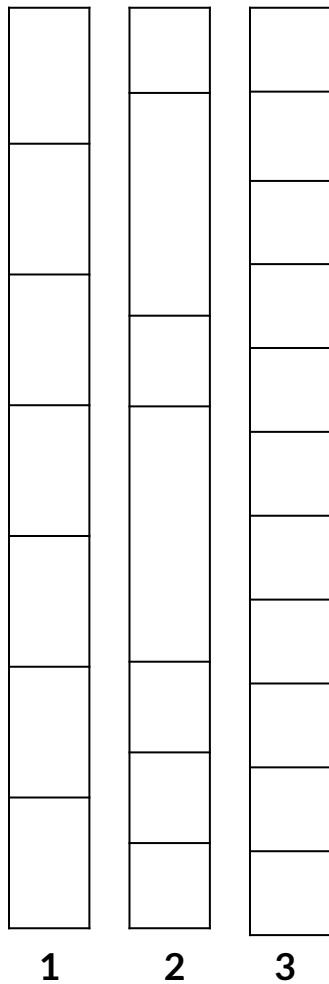


Method

Create a **topic model** of the text
and visualize **topics** over narrative **time**

topics : narrative :: paradigm : syntagm

See if clear textual divisions **emerge**
and **compare** to existing divisions



chap_num	line_num	token_num	token_str	term_str	chunk_id
1	1	0	ARE'	are'	1
1	1	1	U	u	1
1	1	2	XE'	xe'	1
1	1	3	OJER	ojer	1
1	1	4	TZIJ,	tzij	1
1	2	0	Waral	waral	1
1	2	1	K'iche'	k'iche'	1
1	2	2	u	u	1
1	2	3	b'i'.	b'i'	1
1	3	0	WARAL	waral	1

4

chunk_id	a	a'	a'on	ab'aj	ab'ajil	ab'anel	ab'ix	ab'ixik	ab'ixom	achaq
1	0	0	0	1	0	1	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0
4	0	0	0	1	0	0	0	0	0	0
5	0	0	0	2	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0	0	0	0
8	3	0	0	0	0	0	0	0	0	0
9	0	1	0	1	0	0	0	0	0	0
10	0	0	0	1	0	0	0	0	0	0

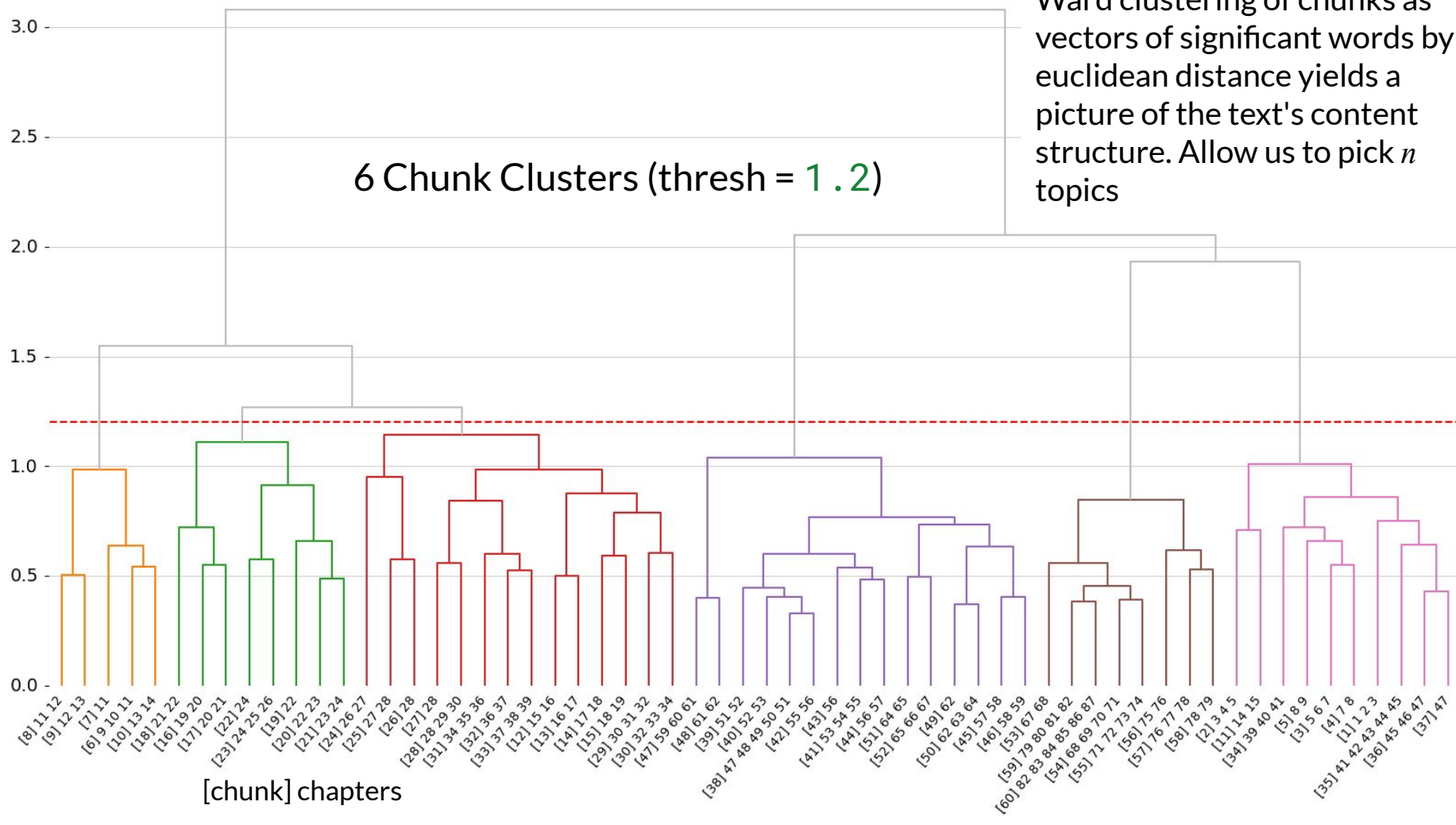
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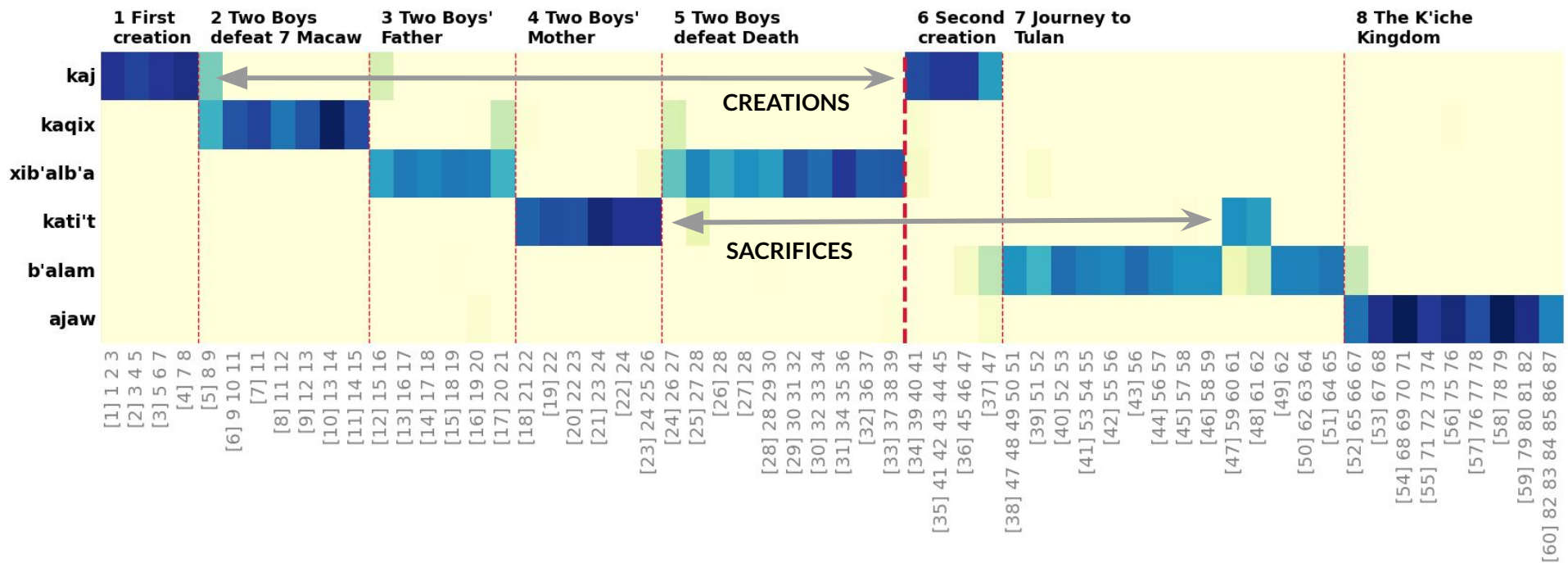
1. Text lines divided by **folio** and **side**
2. Text lines divided by **paragraph** and **chapter**
3. Text lines divided into **equal chunks**
4. Text lines converted into **tokens**
5. Tokens converted into **vectors**

We prepare the text by converting from **folio divisions** into a **vector space** suitable for analysis

6 Chunk Clusters (thresh = 1.2)

Ward clustering of chunks as vectors of significant words by euclidean distance yields a picture of the text's content structure. Allow us to pick n topics





MYTH / GODS

LEGEND / HEROS

HISTORY / KINGS

The first rising of the
Sun, the creation of
corn and the first
humans

Cosmogony

Anthropogony

Sociology